# 1. Setup e Imports

```
import json as _json
import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import Dataset, DataLoader

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from tqdm import tqdm
import os
import sys
from pathlib import Path

# Agregar path del proyecto para imports
sys.path.append(os.path.abspath('../../..'))
from sae.models.sae import SparseAutoencoder
from sae.tools.naming_utils import save_checkpoint, get_model_name, get_extras_id, get_checkp
from sae.tools.experiment_utils import get_experiment_dir, get_plots_dir, get_metrics_dir, ge

# Configuración de visualización para Quarto/PDF
pd.set_option('display.width', 100)
pd.set_option('display.max_columns', None)
np.set_printoptions(linewidth=100, precision=4)
os.environ['COLUMNS'] = '100'

# Configuración de reproducibilidad
torch.manual_seed(42)
np.random.seed(42)

# Device
device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
```

```python
# Print config con formato profesional
config_df = pd.DataFrame({
    'Configuración': ['Device', 'Seed', 'Output Width'],
    'Valor': [str(device), '42', '100']
})

print('\n' + '='*50)
print(' CONFIGURACIÓN INICIAL')
print('='*50)
print(config_df.to_string(index=False))
print('='*50)
```

```
==================================================
 CONFIGURACIÓN INICIAL
==================================================
Configuración Valor
       Device  cuda
         Seed    42
 Output Width   100
==================================================
```

## 2 Metadata para Naming de Checkpoints

```python
# Metadata para naming de checkpoints (separada de hiperparámetros)
NAMING_META = {
    'variante': 'standard',          # Arquitectura del SAE
    'tecnica': 'p-annealing',        # P-annealing (Lp^p con p decreciente)
    'capa': 6,                       # Layer del modelo OthelloGPT
    'juegos': 12500,                # Número de partidas
    'base_path': 'C:\\Users\\Esposa\\Documents\\Repos\\sae-othello-gpt',  # Path base para cl
    'experiment_base_path': os.path.abspath('../../../experiments') ,      # sae/experiments/
    'extras': {
        'expansion': 16,
        'sparsity_coef': 1e-1,
        'epochs': 200,
        'early_stopping': False,
        'learning_rate': 1e-3,
        'p_start': 1.0,
```

```
        'p_end': 0.2,
        'n_sparsity_updates': 10,
        'anneal_start': 5000,
        'batch_size': 2048,
        'warmup_steps': 1000,
        'sparsity_warmup_steps': 2000,
        'estimated_epochs': 200
    }
}

print('\n Metadata de naming:')
for key, value in NAMING_META.items():
    print(f'  {key}: {value}')
```

```
 Metadata de naming:
  variante: standard
  tecnica: p-annealing
  capa: 6
  juegos: 12500
  base_path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt
  experiment_base_path: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments
  extras: {'expansion': 16, 'sparsity_coef': 0.1, 'epochs': 200, 'early_stopping': False, 'le
```

```
extras_id = get_extras_id(
    get_checkpoint_dir(
        NAMING_META['base_path'],
        NAMING_META['variante'],
        NAMING_META['tecnica'],
        NAMING_META['capa'],
        NAMING_META['juegos']
    ),
    NAMING_META['extras']
) if NAMING_META.get('extras') else None

print(f'extras_id: {extras_id}')
```

```
  Nuevo extras registrado: ex2 → {'expansion': 16, 'sparsity_coef': 0.1, 'epochs': 200, 'earl
extras_id: ex2
```

### 3. Configuración del SAE

Las activaciones ya fueron extraídas previamente con `sae/activations/run_extraction.py`. Aquí especificamos qué archivo cargar y los hiperparámetros del SAE.

```
# Configuración del SAE
CONFIG = {
    # Arquitectura
    'input_dim': 512,
    'hidden_dim': 8192,
    'tied_weights': False,

    # Entrenamiento
    'batch_size': 2048,
    'num_epochs': 200,
    'learning_rate': 1e-3,
    'warmup_steps': 1000,        # steps de LR warmup lineal
    'sparsity_warmup_steps': 2000, # steps de rampa de sparsity (0→1)
    'sparsity_coef': 1e-1,       # alpha inicial (loss sum(dim=-1).mean())

    # P-annealing
    'p_start': 1.0,              # p inicial (equivalente a L1)
    'p_end': 0.2,                # p final
    'n_sparsity_updates': 10,    # número de saltos discretos
    'anneal_start': 5000,        # step en que empieza el annealing (después del warmup)
    'estimated_epochs': 200,       # épocas estimadas para condensar el schedule

    # Early Stopping
    'early_stopping': False,
    'patience': 50,                # (inactivo: early_stopping=False)

    # Datos
    'activations_file': Path(f"../../../activations/data/layer{NAMING_META['capa']}_{NAMING_
    'save_dir': Path('./saved_model'),
    'plots_dir': get_plots_dir(
        NAMING_META['experiment_base_path'],
        NAMING_META['variante'],
        NAMING_META['tecnica'],
        NAMING_META['capa'],
        NAMING_META['juegos'],
        extras_id=extras_id
    )
}
```

```python
# Crear directorio local del modelo
CONFIG['save_dir'].mkdir(parents=True, exist_ok=True)

# Mostrar configuración
config_data = {
    'Categoría': [],
    'Parámetro': [],
    'Valor': []
}

for key in ['input_dim', 'hidden_dim', 'tied_weights']:
    config_data['Categoría'].append('Arquitectura')
    config_data['Parámetro'].append(key)
    config_data['Valor'].append(str(CONFIG[key]))

for key in ['batch_size', 'num_epochs', 'learning_rate', 'warmup_steps', 'sparsity_warmup_st
    config_data['Categoría'].append('Entrenamiento')
    config_data['Parámetro'].append(key)
    config_data['Valor'].append(str(CONFIG[key]))

for key in ['p_start', 'p_end', 'n_sparsity_updates', 'anneal_start', 'estimated_epochs']:
    config_data['Categoría'].append('P-Annealing')
    config_data['Parámetro'].append(key)
    config_data['Valor'].append(str(CONFIG[key]))

for key in ['early_stopping', 'patience']:
    config_data['Categoría'].append('Early Stopping')
    config_data['Parámetro'].append(key)
    config_data['Valor'].append(str(CONFIG[key]))

config_table = pd.DataFrame(config_data)

print('\n' + '='*70)
print(' CONFIGURACIÓN DEL SAE')
print('='*70)
print(config_table.to_string(index=False))
print('='*70)

print(f'\nArchivo de activaciones: {CONFIG["activations_file"]}')
print(f'Expansión: {CONFIG["hidden_dim"]/CONFIG["input_dim"]:.1f}x')
print(f'Plots dir: {CONFIG["plots_dir"]}')
print('='*70)
```

```
======================================================================
 CONFIGURACIÓN DEL SAE
======================================================================
     Categoría             Parámetro Valor
  Arquitectura             input_dim   512
  Arquitectura            hidden_dim  8192
  Arquitectura          tied_weights False
 Entrenamiento            batch_size  2048
 Entrenamiento            num_epochs   200
 Entrenamiento         learning_rate 0.001
 Entrenamiento          warmup_steps  1000
 Entrenamiento sparsity_warmup_steps  2000
 Entrenamiento         sparsity_coef   0.1
   P-Annealing               p_start   1.0
   P-Annealing                 p_end   0.2
   P-Annealing    n_sparsity_updates    10
   P-Annealing          anneal_start  5000
   P-Annealing      estimated_epochs   200
Early Stopping        early_stopping False
Early Stopping              patience    50
======================================================================

Archivo de activaciones: ..\..\..\activations\data\layer6_12500games.npy
Expansión: 16.0x
Plots dir: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standa
annealing_standard\12500games\ex2\plots
======================================================================
```

### 4. Dataset de Activaciones (Train/Validation Split 80-20%)

```python
class ActivationsDataset(Dataset):
    """Dataset para cargar activaciones extraídas"""

    def __init__(self, activations_path):
        self.activations = np.load(activations_path)

    def __len__(self):
```

```
        return len(self.activations)

    def __getitem__(self, idx):
        return torch.tensor(self.activations[idx], dtype=torch.float32)

# Cargar dataset completo
full_dataset = ActivationsDataset(CONFIG['activations_file'])

# Información del dataset con formato profesional
dataset_info = pd.DataFrame({
    'Propiedad': ['Total muestras', 'Shape', 'Dtype', 'Tamaño en memoria'],
    'Valor': [
        f"{len(full_dataset):,}",
        str(full_dataset.activations.shape),
        str(full_dataset.activations.dtype),
        f"{full_dataset.activations.nbytes / 1024**2:.2f} MB"
    ]
})
print('\n' + '='*50)
print(' DATASET DE ACTIVACIONES')
print('='*50)
print(dataset_info.to_string(index=False))
print('='*50)

# Split train/validation (80-20)
train_size = int(0.8 * len(full_dataset))
val_size = len(full_dataset) - train_size
train_dataset, val_dataset = torch.utils.data.random_split(
    full_dataset,
    [train_size, val_size],
    generator=torch.Generator().manual_seed(42)
)

# Resumen del split con formato profesional
split_info = pd.DataFrame({
    'Conjunto': ['Train', 'Validation', 'Total'],
    'Muestras': [
        f"{len(train_dataset):,}",
        f"{len(val_dataset):,}",
        f"{len(full_dataset):,}"
    ],
    'Porcentaje': [
```

```
        f"{len(train_dataset)/len(full_dataset):.1%}",
        f"{len(val_dataset)/len(full_dataset):.1%}",
        "100.0%"
    ]
})
print('\n' + '='*50)
print(' SPLIT DE DATOS (80-20)')
print('='*50)
print(split_info.to_string(index=False))
print('='*50)

# Crear DataLoaders
train_loader = DataLoader(
    train_dataset,
    batch_size=CONFIG['batch_size'],
    shuffle=True,
    num_workers=0,
    pin_memory=(device.type == 'cuda')
)

val_loader = DataLoader(
    val_dataset,
    batch_size=CONFIG['batch_size'],
    shuffle=False,  # No shuffle para validación
    num_workers=0,
    pin_memory=(device.type == 'cuda')
)

# Resumen de DataLoaders
loader_info = pd.DataFrame({
    'DataLoader': ['Train', 'Validation'],
    'Batches': [len(train_loader), len(val_loader)],
    'Batch Size': [CONFIG['batch_size'], CONFIG['batch_size']],
    'Shuffle': ['Sí', 'No']
})

print('\n' + '='*50)
print(' DATALOADERS')
print('='*50)
print(loader_info.to_string(index=False))
print('='*50)
```

```
================================================
 DATASET DE ACTIVACIONES
================================================
        Propiedad         Valor
   Total muestras       737,500
            Shape (737500, 512)
            Dtype       float32
Tamaño en memoria    1440.43 MB
================================================

================================================
 SPLIT DE DATOS (80-20)
================================================
  Conjunto Muestras Porcentaje
     Train  590,000      80.0%
Validation  147,500      20.0%
     Total  737,500     100.0%
================================================

================================================
 DATALOADERS
================================================
DataLoader  Batches  Batch Size Shuffle
     Train      289        2048      Sí
Validation       73        2048      No
================================================
```

### 5. Modelo: Sparse Autoencoder

Modelo importado desde `sae.models.sae.SparseAutoencoder`

**Arquitectura:**

- **Encoder**: Linear → ReLU (sparsity natural)- **Decoder**: Linear (reconstrucción)

```
# Crear modelo desde el módulo
model = SparseAutoencoder(
    input_dim=CONFIG['input_dim'],
    hidden_dim=CONFIG['hidden_dim'],
    tied_weights=CONFIG['tied_weights']
).to(device)
```

```python
# Calcular parámetros
num_params = sum(p.numel() for p in model.parameters())
encoder_params = sum(p.numel() for p in model.encoder.parameters())
if CONFIG['tied_weights']:
    decoder_params = CONFIG['input_dim']  # Solo bias
else:
    decoder_params = sum(p.numel() for p in model.decoder.parameters())

# Resumen del modelo con formato profesional
model_info = pd.DataFrame({
    'Componente': ['Input', 'Encoder', 'Hidden (Latent)', 'Decoder', 'Output', 'Total'],
    'Dimensión': [
        CONFIG['input_dim'],
        f"{CONFIG['input_dim']} → {CONFIG['hidden_dim']}",
        CONFIG['hidden_dim'],
        f"{CONFIG['hidden_dim']} → {CONFIG['input_dim']}",
        CONFIG['input_dim'],
        '-'
    ],
    'Parámetros': [
        '-',
        f"{encoder_params:,}",
        '-',
        f"{decoder_params:,}",
        '-',
        f"{num_params:,}"
    ]
})

print('\n' + '='*70)
print(' ARQUITECTURA DEL MODELO')
print('='*70)
print(model_info.to_string(index=False))
print('='*70)
print(f'\n Factor de expansión: {CONFIG["hidden_dim"]/CONFIG["input_dim"]:.1f}x')
print(f' Tied weights: {"Sí" if CONFIG["tied_weights"] else "No"}')
print(f' Device: {device}')
print('='*70)
```

```
======================================================================
 ARQUITECTURA DEL MODELO
```

```
======================================================================
     Componente  Dimensión Parámetros
          Input        512          -
        Encoder 512 → 8192  4,202,496
Hidden (Latent)       8192          -
        Decoder 8192 → 512  4,194,304
         Output        512          -
          Total          -  8,397,312
======================================================================

 Factor de expansión: 16.0x
 Tied weights: No
 Device: cuda
======================================================================
```

### 6. Función de Pérdida

$$\textbf{Loss} = \textbf{MSE(reconstrucción)} + \quad \times \textbf{L1(activaciones)}$$

- **MSE**: Mide qué tan bien reconstruimos las activaciones originales
- **L1**: Penaliza activaciones grandes → fuerza sparsity

```python
class ConstrainedAdam(torch.optim.Adam):
    """
    Adam donde las columnas de decoder se proyectan a norma unitaria en cada step.
    Antes del paso: elimina la componente del gradiente paralela al decoder.
    Después del paso: renormaliza las columnas del decoder a norma 1.
    """
    def __init__(self, params, constrained_params, lr, betas=(0.9, 0.999)):
        super().__init__(params, lr=lr, betas=betas)
        self.constrained_params = list(constrained_params)

    def step(self, closure=None):
        with torch.no_grad():
            for p in self.constrained_params:
                normed_p = p / p.norm(dim=0, keepdim=True)
                p.grad -= (p.grad * normed_p).sum(dim=0, keepdim=True) * normed_p
        super().step(closure=closure)
        with torch.no_grad():
            for p in self.constrained_params:
                p /= p.norm(dim=0, keepdim=True)
```

```
def loss_function(x, reconstruction, hidden, sparsity_coeff, p, step=0, sparsity_warmup_steps
    """
    Pérdida compuesta: MSE (sum/mean) + Lp^p penalty con sparsity warmup

    Args:
        x: Activaciones originales
        reconstruction: Activaciones reconstruidas
        hidden: Activaciones latentes (sparse, >= 0 por ReLU)
        sparsity_coeff: Peso adaptativo de la penalización (alpha)
        p: Exponente actual del schedule de annealing
        step: Step global actual (para sparsity warmup)
        sparsity_warmup_steps: Steps de rampa de sparsity (0 = desactivado)
    """
    # BUG2 FIX: sum sobre features luego mean sobre batch (igual que referencia)
    mse_loss = (reconstruction - x).pow(2).sum(dim=-1).mean()
    # BUG4 FIX: sparsity warmup - rampa lineal de 0 a 1
    sparsity_scale = min(step / sparsity_warmup_steps, 1.0) if sparsity_warmup_steps > 0 else
    sparsity_loss = hidden.pow(p).sum(dim=1).mean()   # Lp^p: Σ|f_i|^p
    total_loss = mse_loss + sparsity_coeff * sparsity_scale * sparsity_loss
    return total_loss, mse_loss, sparsity_loss


# ConstrainedAdam: pasa model.decoder.parameters() como parámetros constrained
optimizer = ConstrainedAdam(
    model.parameters(),
    model.decoder.parameters(),
    lr=CONFIG['learning_rate']
)

# BUG3 FIX: LR warmup lineal - sube de 0 a lr en warmup_steps
def _lr_lambda(step):
    warmup = CONFIG.get('warmup_steps', 1000)
    return min(step / warmup, 1.0) if warmup > 0 else 1.0

scheduler = torch.optim.lr_scheduler.LambdaLR(optimizer, lr_lambda=_lr_lambda)

optim_info = pd.DataFrame({
    'Parámetro': ['Optimizador', 'Learning Rate', 'Sparsity Coef (inicial)', 'p_start', 'p_en
    'Valor': [
        'ConstrainedAdam',
        f"{CONFIG['learning_rate']:.0e}",
        f"{CONFIG['sparsity_coef']:.0e}",
```

```
        f"{CONFIG['p_start']}",
        f"{CONFIG['p_end']}",
        f"{CONFIG['n_sparsity_updates']}",
        f"{CONFIG['anneal_start']}"
    ]
})

print('\n' + '='*55)
print(' OPTIMIZACIÓN')
print('='*55)
print(optim_info.to_string(index=False))
print('='*55)
```

```
=======================================================
 OPTIMIZACIÓN
=======================================================
             Parámetro           Valor
           Optimizador ConstrainedAdam
         Learning Rate           1e-03
Sparsity Coef (inicial)          1e-01
               p_start            1.0
                 p_end            0.2
    n_sparsity_updates             10
          anneal_start           5000
=======================================================
```

### 7. Loop de Entrenamiento

```
# Historial de entrenamiento
history = {
    'train_total_loss': [],
    'train_mse_loss': [],
    'train_sparsity_loss': [],
    'train_l0_sparsity': [],
    'train_fraction_active': [],
    'val_total_loss': [],
    'val_mse_loss': [],
    'val_sparsity_loss': [],
    'val_l0_sparsity': [],
```

```
    'val_fraction_active': [],
    'sparsity_coef_history': [],   # snapshot de sparsity_coeff al final de cada época
    'p_history': [],               # snapshot de p al final de cada época
}

# Early stopping variables
best_val_mse = float('inf')
best_epoch = 0
epochs_no_improve = 0

# ========== P-ANNEALING STATE ==========
p = CONFIG['p_start']
next_p = None
p_step_count = 0
sparsity_coeff = CONFIG['sparsity_coef']
sparsity_queue = []
sparsity_queue_length = 10

total_steps = CONFIG['num_epochs'] * len(train_loader)
# BUG5 FIX: usar expected_steps (no total_steps) para que el schedule
# de annealing complete antes del early stopping
estimated_epochs = CONFIG.get('estimated_epochs', 50)
expected_steps = estimated_epochs * len(train_loader)
_update_steps_tensor = torch.linspace(
    CONFIG['anneal_start'], expected_steps, CONFIG['n_sparsity_updates'], dtype=torch.int
)
sparsity_update_steps_set = set(_update_steps_tensor.tolist())
sparsity_update_steps_list = sorted(sparsity_update_steps_set)
p_values = torch.linspace(CONFIG['p_start'], CONFIG['p_end'], CONFIG['n_sparsity_updates']).t
global_step = 0

print(f'\n Iniciando entrenamiento: {CONFIG["num_epochs"]} épocas máximo')
print(f' P-annealing: p {CONFIG["p_start"]} → {CONFIG["p_end"]} en {CONFIG["n_sparsity_update
print(f'   Early stopping: patience={CONFIG["patience"]} épocas')
print('=' * 70)

for epoch in range(CONFIG['num_epochs']):

    # ========== ENTRENAMIENTO ==========
    model.train()

    epoch_total_loss = 0
```

```
    epoch_mse_loss = 0
    epoch_sparsity_loss = 0
    epoch_l0 = 0
    epoch_fraction_active = 0

    pbar = tqdm(train_loader, desc=f'Época {epoch+1}/{CONFIG["num_epochs"]} [Train] p={p:.3f}

    for batch in pbar:
        x = batch.to(device)

        # Forward pass
        reconstruction, hidden = model(x)

        # ---- P-ANNEALING: lookahead para la cola ----
        with torch.no_grad():
            if next_p is not None:
                lp_curr = hidden.pow(p).sum(dim=1).mean().item()
                lp_next = hidden.pow(next_p).sum(dim=1).mean().item()
                sparsity_queue.append([lp_curr, lp_next])
                sparsity_queue = sparsity_queue[-sparsity_queue_length:]

            # Verificar si este step es un punto de actualización de p
            if (global_step in sparsity_update_steps_set
                    and p_step_count < CONFIG['n_sparsity_updates']
                    and global_step >= sparsity_update_steps_list[p_step_count]):
                # Re-escalar sparsity_coeff para compensar el cambio de magnitud de Lp^p
                if next_p is not None and len(sparsity_queue) > 0:
                    local_curr = sum(i[0] for i in sparsity_queue) / len(sparsity_queue)
                    local_next = sum(i[1] for i in sparsity_queue) / len(sparsity_queue)
                    if local_next > 0:
                        sparsity_coeff = sparsity_coeff * (local_curr / local_next)
                # Actualizar p al siguiente valor del schedule
                p = p_values[p_step_count]
                if p_step_count < CONFIG['n_sparsity_updates'] - 1:
                    next_p = p_values[p_step_count + 1]
                else:
                    next_p = CONFIG['p_end']
                p_step_count += 1

        # Calcular pérdida con p actual
        total_loss, mse_loss, sparsity_loss = loss_function(
            x, reconstruction, hidden, sparsity_coeff, p,
```

```
            step=global_step,
            sparsity_warmup_steps=CONFIG.get('sparsity_warmup_steps', 0)
        )

        # Backward pass
        optimizer.zero_grad()
        total_loss.backward()
        optimizer.step()
        scheduler.step()  # BUG3 FIX: LR warmup step

        global_step += 1

        # Métricas
        with torch.no_grad():
            l0 = (hidden > 0).float().sum(dim=1).mean().item()
            fraction_active = (hidden > 0).float().mean().item()

        epoch_total_loss += total_loss.item()
        epoch_mse_loss += mse_loss.item()
        epoch_sparsity_loss += sparsity_loss.item()
        epoch_l0 += l0
        epoch_fraction_active += fraction_active

        pbar.set_postfix({
            'loss': f'{total_loss.item():.4f}',
            'mse': f'{mse_loss.item():.4f}',
            'l0': f'{l0:.1f}',
            'p': f'{p:.3f}'
        })

    # Promedios de entrenamiento
    num_train_batches = len(train_loader)
    avg_train_total_loss = epoch_total_loss / num_train_batches
    avg_train_mse_loss = epoch_mse_loss / num_train_batches
    avg_train_sparsity_loss = epoch_sparsity_loss / num_train_batches
    avg_train_l0 = epoch_l0 / num_train_batches
    avg_train_fraction_active = epoch_fraction_active / num_train_batches

    # ========== VALIDACIÓN ==========
    model.eval()

    val_total_loss = 0
```

```
    val_mse_loss = 0
    val_sparsity_loss = 0
    val_l0 = 0
    val_fraction_active = 0

    with torch.no_grad():
        for batch in val_loader:
            x = batch.to(device)
            reconstruction, hidden = model(x)
            total_loss, mse_loss, sparsity_loss = loss_function(
                x, reconstruction, hidden, sparsity_coeff, p,
                step=global_step,
                sparsity_warmup_steps=CONFIG.get('sparsity_warmup_steps', 0)
            )
            l0 = (hidden > 0).float().sum(dim=1).mean().item()
            fraction_active = (hidden > 0).float().mean().item()

            val_total_loss += total_loss.item()
            val_mse_loss += mse_loss.item()
            val_sparsity_loss += sparsity_loss.item()
            val_l0 += l0
            val_fraction_active += fraction_active

    num_val_batches = len(val_loader)
    avg_val_total_loss = val_total_loss / num_val_batches
    avg_val_mse_loss = val_mse_loss / num_val_batches
    avg_val_sparsity_loss = val_sparsity_loss / num_val_batches
    avg_val_l0 = val_l0 / num_val_batches
    avg_val_fraction_active = val_fraction_active / num_val_batches

    # Guardar en historial
    history['train_total_loss'].append(avg_train_total_loss)
    history['train_mse_loss'].append(avg_train_mse_loss)
    history['train_sparsity_loss'].append(avg_train_sparsity_loss)
    history['train_l0_sparsity'].append(avg_train_l0)
    history['train_fraction_active'].append(avg_train_fraction_active)

    history['val_total_loss'].append(avg_val_total_loss)
    history['val_mse_loss'].append(avg_val_mse_loss)
    history['val_sparsity_loss'].append(avg_val_sparsity_loss)
    history['val_l0_sparsity'].append(avg_val_l0)
    history['val_fraction_active'].append(avg_val_fraction_active)
```

```
    history['sparsity_coef_history'].append(sparsity_coeff)
    history['p_history'].append(p)

    # ========== EARLY STOPPING ==========
    if CONFIG['early_stopping']:
        if avg_val_mse_loss < best_val_mse:
            best_val_mse = avg_val_mse_loss
            best_epoch = epoch + 1
            epochs_no_improve = 0

            best_checkpoint = {
                'model_state_dict': model.state_dict(),
                'optimizer_state_dict': optimizer.state_dict(),
                'config': CONFIG,
                'history': history,
                'epoch': best_epoch,
                'val_mse': best_val_mse
            }

            best_model_name = get_model_name(
                variante=NAMING_META['variante'],
                tecnica=NAMING_META['tecnica'],
                capa=NAMING_META['capa'],
                juegos=NAMING_META['juegos'],
                estado='best'
            )
            torch.save(best_checkpoint, CONFIG['save_dir'] / best_model_name)

            save_checkpoint(
                model,
                base_path=NAMING_META['base_path'],
                variante=NAMING_META['variante'],
                tecnica=NAMING_META['tecnica'],
                capa=NAMING_META['capa'],
                juegos=NAMING_META['juegos'],
                estado='best',
                extras=NAMING_META['extras'],
                config=CONFIG,
                history=history,
                optimizer=optimizer
            )
```

```
            print(f' Mejor modelo guardado en época {best_epoch} (Val MSE: {best_val_mse:.4f}
        else:
            epochs_no_improve += 1
            if epochs_no_improve >= CONFIG['patience']:
                print(f'\n  Early stopping activado en época {epoch+1}')
                print(f'   Mejor modelo: época {best_epoch} con Val MSE = {best_val_mse:.4f}
                break

    # Log cada 5 épocas
    if (epoch + 1) % 5 == 0 or epoch == 0:
        print(f'\n Época {epoch+1}/{CONFIG["num_epochs"]} | p: {p:.3f} |  : {sparsity_coeff:.
        print(f'  Train - Total: {avg_train_total_loss:.4f} | MSE: {avg_train_mse_loss:.4f}
        print(f'  Val   - Total: {avg_val_total_loss:.4f} | MSE: {avg_val_mse_loss:.4f} | L0
        if CONFIG['early_stopping']:
            print(f'   Best Val MSE: {best_val_mse:.4f} @ época {best_epoch} | Sin mejora: {e

print('\n' + '='*70)
print(' RESUMEN DEL ENTRENAMIENTO')
print('='*70)

if CONFIG['early_stopping']:
    print(f'Mejor modelo en época {best_epoch} con Val MSE = {best_val_mse:.4f}')
    print(f'Early stopping tras {len(history["train_total_loss"])} épocas')
else:
    print(f'Entrenamiento completo: {CONFIG["num_epochs"]} épocas')

final_metrics = pd.DataFrame({
    'Métrica': ['Total Loss', 'MSE Loss', 'L0 Sparsity', 'Fraction Active'],
    'Train': [
        f"{history['train_total_loss'][-1]:.4f}",
        f"{history['train_mse_loss'][-1]:.4f}",
        f"{history['train_l0_sparsity'][-1]:.1f}",
        f"{history['train_fraction_active'][-1]:.2%}"
    ],
    'Validation': [
        f"{history['val_total_loss'][-1]:.4f}",
        f"{history['val_mse_loss'][-1]:.4f}",
        f"{history['val_l0_sparsity'][-1]:.1f}",
        f"{history['val_fraction_active'][-1]:.2%}"
    ]
})
```

```
print('\nMétricas Finales (Última Época):')
print(final_metrics.to_string(index=False))
print('='*70)
```

```
 Iniciando entrenamiento: 200 épocas máximo
 P-annealing: p 1.0 → 0.2 en 10 saltos
  Early stopping: patience=50 épocas
======================================================================

Época 1/200 [Train] p=1.000  =1.00e-01: 100%|      | 289/289 [00:10<00:00, 27.21it/s, loss=9.0


 Época 1/200 | p: 1.000 |  : 1.00e-01
  Train - Total: 12.6973 | MSE: 9.3549 | L0: 5165.7
  Val   - Total: 9.1849 | MSE: 2.5578 | L0: 5509.9

Época 2/200 [Train] p=1.000  =1.00e-01: 100%|      | 289/289 [00:10<00:00, 27.87it/s, loss=13.
Época 3/200 [Train] p=1.000  =1.00e-01: 100%|      | 289/289 [00:10<00:00, 28.15it/s, loss=18.
Época 4/200 [Train] p=1.000  =1.00e-01: 100%|      | 289/289 [00:10<00:00, 27.59it/s, loss=21.
Época 5/200 [Train] p=1.000  =1.00e-01: 100%|      | 289/289 [00:10<00:00, 27.95it/s, loss=23.


 Época 5/200 | p: 1.000 |  : 1.00e-01
  Train - Total: 21.8012 | MSE: 5.9863 | L0: 3044.8
  Val   - Total: 23.4716 | MSE: 6.5202 | L0: 2893.9

Época 6/200 [Train] p=1.000  =1.00e-01: 100%|      | 289/289 [00:10<00:00, 27.93it/s, loss=25.
Época 7/200 [Train] p=1.000  =1.00e-01: 100%|      | 289/289 [00:10<00:00, 27.97it/s, loss=27.
Época 8/200 [Train] p=1.000  =1.00e-01: 100%|      | 289/289 [00:10<00:00, 26.95it/s, loss=27.
Época 9/200 [Train] p=1.000  =1.00e-01: 100%|      | 289/289 [00:10<00:00, 26.76it/s, loss=26.
Época 10/200 [Train] p=1.000  =1.00e-01: 100%|     | 289/289 [00:10<00:00, 27.78it/s, loss=26


 Época 10/200 | p: 1.000 |  : 1.00e-01
  Train - Total: 26.3320 | MSE: 6.2451 | L0: 2004.6
  Val   - Total: 26.5353 | MSE: 6.3282 | L0: 1979.0
```

```
Época 11/200 [Train] p=1.000  =1.00e-01: 100%|      | 289/289 [00:11<00:00, 24.79it/s, loss=25
Época 12/200 [Train] p=1.000  =1.00e-01: 100%|      | 289/289 [00:12<00:00, 23.59it/s, loss=25
Época 13/200 [Train] p=1.000  =1.00e-01: 100%|      | 289/289 [00:11<00:00, 26.17it/s, loss=24
Época 14/200 [Train] p=1.000  =1.00e-01: 100%|      | 289/289 [00:10<00:00, 27.88it/s, loss=25
Época 15/200 [Train] p=1.000  =1.00e-01: 100%|      | 289/289 [00:10<00:00, 27.75it/s, loss=24


 Época 15/200 | p: 1.000 |  : 1.00e-01
  Train - Total: 25.1189 | MSE: 4.7745 | L0: 1582.6
  Val   - Total: 25.5201 | MSE: 5.0776 | L0: 1566.3

Época 16/200 [Train] p=1.000  =1.00e-01: 100%|      | 289/289 [00:11<00:00, 26.14it/s, loss=24
Época 17/200 [Train] p=1.000  =1.00e-01: 100%|      | 289/289 [00:12<00:00, 23.67it/s, loss=25
Época 18/200 [Train] p=1.000  =1.00e-01: 100%|      | 289/289 [00:10<00:00, 26.73it/s, loss=24
Época 19/200 [Train] p=1.000  =1.00e-01: 100%|      | 289/289 [00:11<00:00, 25.98it/s, loss=24
Época 20/200 [Train] p=1.000  =1.00e-01: 100%|      | 289/289 [00:11<00:00, 25.55it/s, loss=25


 Época 20/200 | p: 1.000 |  : 1.00e-01
  Train - Total: 24.9290 | MSE: 4.3603 | L0: 1283.0
  Val   - Total: 25.3742 | MSE: 4.7392 | L0: 1270.2

Época 21/200 [Train] p=1.000  =1.00e-01: 100%|      | 289/289 [00:13<00:00, 21.91it/s, loss=24
Época 22/200 [Train] p=1.000  =1.00e-01: 100%|      | 289/289 [00:12<00:00, 23.43it/s, loss=25
Época 23/200 [Train] p=1.000  =1.00e-01: 100%|      | 289/289 [00:12<00:00, 23.18it/s, loss=25
Época 24/200 [Train] p=1.000  =1.00e-01: 100%|      | 289/289 [00:12<00:00, 23.25it/s, loss=25
Época 25/200 [Train] p=1.000  =1.00e-01: 100%|      | 289/289 [00:12<00:00, 22.99it/s, loss=24


 Época 25/200 | p: 1.000 |  : 1.00e-01
  Train - Total: 24.7409 | MSE: 3.8364 | L0: 1060.6
  Val   - Total: 25.1585 | MSE: 4.1246 | L0: 1055.7

Época 26/200 [Train] p=1.000  =1.00e-01: 100%|      | 289/289 [00:12<00:00, 22.83it/s, loss=24
Época 27/200 [Train] p=1.000  =1.00e-01: 100%|      | 289/289 [00:12<00:00, 23.10it/s, loss=24
Época 28/200 [Train] p=1.000  =1.00e-01: 100%|      | 289/289 [00:12<00:00, 23.03it/s, loss=24
Época 29/200 [Train] p=1.000  =1.00e-01: 100%|      | 289/289 [00:12<00:00, 23.47it/s, loss=24
Época 30/200 [Train] p=1.000  =1.00e-01: 100%|      | 289/289 [00:12<00:00, 23.19it/s, loss=25
```

```
 Época 30/200 | p: 1.000 |  : 1.00e-01
  Train - Total: 24.5888 | MSE: 3.3561 | L0: 901.9
  Val   - Total: 24.9975 | MSE: 3.6278 | L0: 895.0

Época 31/200 [Train] p=1.000  =1.00e-01: 100%|     | 289/289 [00:12<00:00, 23.19it/s, loss=24
Época 32/200 [Train] p=1.000  =1.00e-01: 100%|     | 289/289 [00:13<00:00, 22.16it/s, loss=24
Época 33/200 [Train] p=1.000  =1.00e-01: 100%|     | 289/289 [00:12<00:00, 22.81it/s, loss=24
Época 34/200 [Train] p=1.000  =1.00e-01: 100%|     | 289/289 [00:12<00:00, 23.20it/s, loss=24
Época 35/200 [Train] p=1.000  =1.00e-01: 100%|     | 289/289 [00:12<00:00, 22.43it/s, loss=24


 Época 35/200 | p: 1.000 |  : 1.00e-01
  Train - Total: 24.4751 | MSE: 2.9603 | L0: 788.7
  Val   - Total: 24.8204 | MSE: 3.1782 | L0: 782.8

Época 36/200 [Train] p=1.000  =1.00e-01: 100%|     | 289/289 [00:12<00:00, 23.60it/s, loss=24
Época 37/200 [Train] p=1.000  =1.00e-01: 100%|     | 289/289 [00:13<00:00, 22.15it/s, loss=24
Época 38/200 [Train] p=1.000  =1.00e-01: 100%|     | 289/289 [00:12<00:00, 22.83it/s, loss=24
Época 39/200 [Train] p=0.911  =9.37e-02: 100%|     | 289/289 [00:12<00:00, 22.96it/s, loss=24
Época 40/200 [Train] p=0.911  =9.37e-02: 100%|     | 289/289 [00:12<00:00, 23.17it/s, loss=23


 Época 40/200 | p: 0.911 |  : 9.37e-02
  Train - Total: 24.0645 | MSE: 2.1056 | L0: 630.4
  Val   - Total: 24.3170 | MSE: 2.2802 | L0: 624.3

Época 41/200 [Train] p=0.911  =9.37e-02: 100%|     | 289/289 [00:12<00:00, 23.37it/s, loss=24
Época 42/200 [Train] p=0.911  =9.37e-02: 100%|     | 289/289 [00:12<00:00, 23.24it/s, loss=23
Época 43/200 [Train] p=0.911  =9.37e-02: 100%|     | 289/289 [00:12<00:00, 22.66it/s, loss=24
Época 44/200 [Train] p=0.911  =9.37e-02: 100%|     | 289/289 [00:12<00:00, 23.00it/s, loss=23
Época 45/200 [Train] p=0.911  =9.37e-02: 100%|     | 289/289 [00:12<00:00, 22.89it/s, loss=24


 Época 45/200 | p: 0.911 |  : 9.37e-02
  Train - Total: 23.9648 | MSE: 1.9820 | L0: 581.4
  Val   - Total: 24.1969 | MSE: 2.1064 | L0: 581.2
```

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Época 46/200 [Train] p=0.911  =9.37e-02: 100%|      | 289/289 [00:12<00:00, 23.10it/s, loss=24
Época 47/200 [Train] p=0.911  =9.37e-02: 100%|      | 289/289 [00:12<00:00, 23.46it/s, loss=23
Época 48/200 [Train] p=0.911  =9.37e-02: 100%|      | 289/289 [00:12<00:00, 22.59it/s, loss=24
Época 49/200 [Train] p=0.911  =9.37e-02: 100%|      | 289/289 [00:12<00:00, 23.47it/s, loss=23
Época 50/200 [Train] p=0.911  =9.37e-02: 100%|      | 289/289 [00:12<00:00, 22.94it/s, loss=23


 Época 50/200 | p: 0.911 |  : 9.37e-02
  Train - Total: 23.8838 | MSE: 1.8861 | L0: 549.4
  Val   - Total: 24.0797 | MSE: 1.9807 | L0: 549.1

Época 51/200 [Train] p=0.911  =9.37e-02: 100%|      | 289/289 [00:12<00:00, 22.68it/s, loss=23
Época 52/200 [Train] p=0.911  =9.37e-02: 100%|      | 289/289 [00:12<00:00, 23.17it/s, loss=23
Época 53/200 [Train] p=0.911  =9.37e-02: 100%|      | 289/289 [00:12<00:00, 22.54it/s, loss=23
Época 54/200 [Train] p=0.911  =9.37e-02: 100%|      | 289/289 [00:12<00:00, 22.63it/s, loss=24
Época 55/200 [Train] p=0.911  =9.37e-02: 100%|      | 289/289 [00:12<00:00, 22.90it/s, loss=24


 Época 55/200 | p: 0.911 |  : 9.37e-02
  Train - Total: 23.8234 | MSE: 1.7951 | L0: 522.9
  Val   - Total: 23.9952 | MSE: 1.8564 | L0: 524.3

Época 56/200 [Train] p=0.911  =9.37e-02: 100%|      | 289/289 [00:12<00:00, 22.77it/s, loss=24
Época 57/200 [Train] p=0.911  =9.37e-02: 100%|      | 289/289 [00:12<00:00, 22.77it/s, loss=23
Época 58/200 [Train] p=0.911  =9.37e-02: 100%|      | 289/289 [00:12<00:00, 23.65it/s, loss=23
Época 59/200 [Train] p=0.822  =8.99e-02: 100%|      | 289/289 [00:13<00:00, 22.03it/s, loss=23
Época 60/200 [Train] p=0.822  =8.99e-02: 100%|      | 289/289 [00:12<00:00, 22.88it/s, loss=23


 Época 60/200 | p: 0.822 |  : 8.99e-02
  Train - Total: 23.6976 | MSE: 1.6801 | L0: 489.0
  Val   - Total: 23.8504 | MSE: 1.7364 | L0: 489.0

Época 61/200 [Train] p=0.822  =8.99e-02: 100%|      | 289/289 [00:12<00:00, 22.99it/s, loss=23
Época 62/200 [Train] p=0.822  =8.99e-02: 100%|      | 289/289 [00:12<00:00, 22.73it/s, loss=24
Época 63/200 [Train] p=0.822  =8.99e-02: 100%|      | 289/289 [00:12<00:00, 22.87it/s, loss=23
Época 64/200 [Train] p=0.822  =8.99e-02: 100%|      | 289/289 [00:12<00:00, 22.77it/s, loss=23
Época 65/200 [Train] p=0.822  =8.99e-02: 100%|      | 289/289 [00:12<00:00, 22.88it/s, loss=23
```

```
 Época 65/200 | p: 0.822 |  : 8.99e-02
  Train - Total: 23.5719 | MSE: 1.5788 | L0: 469.3
  Val   - Total: 23.7337 | MSE: 1.6669 | L0: 470.6

Época 66/200 [Train] p=0.822  =8.99e-02: 100%|     | 289/289 [00:12<00:00, 22.75it/s, loss=23
Época 67/200 [Train] p=0.822  =8.99e-02: 100%|     | 289/289 [00:12<00:00, 22.76it/s, loss=23
Época 68/200 [Train] p=0.822  =8.99e-02: 100%|     | 289/289 [00:12<00:00, 22.84it/s, loss=23
Época 69/200 [Train] p=0.822  =8.99e-02: 100%|     | 289/289 [00:12<00:00, 22.94it/s, loss=23
Época 70/200 [Train] p=0.822  =8.99e-02: 100%|     | 289/289 [00:12<00:00, 23.12it/s, loss=23


 Época 70/200 | p: 0.822 |  : 8.99e-02
  Train - Total: 23.5149 | MSE: 1.5067 | L0: 459.7
  Val   - Total: 23.6674 | MSE: 1.5922 | L0: 460.6

Época 71/200 [Train] p=0.822  =8.99e-02: 100%|     | 289/289 [00:12<00:00, 22.91it/s, loss=23
Época 72/200 [Train] p=0.822  =8.99e-02: 100%|     | 289/289 [00:12<00:00, 22.46it/s, loss=23
Época 73/200 [Train] p=0.822  =8.99e-02: 100%|     | 289/289 [00:12<00:00, 22.92it/s, loss=23
Época 74/200 [Train] p=0.822  =8.99e-02: 100%|     | 289/289 [00:12<00:00, 22.89it/s, loss=24
Época 75/200 [Train] p=0.822  =8.99e-02: 100%|     | 289/289 [00:12<00:00, 23.14it/s, loss=23


 Época 75/200 | p: 0.822 |  : 8.99e-02
  Train - Total: 23.4891 | MSE: 1.4578 | L0: 455.5
  Val   - Total: 23.6461 | MSE: 1.5163 | L0: 458.4

Época 76/200 [Train] p=0.822  =8.99e-02: 100%|     | 289/289 [00:12<00:00, 23.75it/s, loss=23
Época 77/200 [Train] p=0.822  =8.99e-02: 100%|     | 289/289 [00:12<00:00, 22.72it/s, loss=23
Época 78/200 [Train] p=0.822  =8.99e-02: 100%|     | 289/289 [00:12<00:00, 22.89it/s, loss=23
Época 79/200 [Train] p=0.822  =8.99e-02: 100%|     | 289/289 [00:12<00:00, 23.09it/s, loss=23
Época 80/200 [Train] p=0.733  =8.67e-02: 100%|     | 289/289 [00:12<00:00, 22.70it/s, loss=23


 Época 80/200 | p: 0.733 |  : 8.67e-02
  Train - Total: 23.4633 | MSE: 1.4479 | L0: 447.8
  Val   - Total: 23.6101 | MSE: 1.5425 | L0: 448.5
```

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Época 81/200 [Train] p=0.733  =8.67e-02: 100%|      | 289/289 [00:12<00:00, 22.88it/s, loss=23
Época 82/200 [Train] p=0.733  =8.67e-02: 100%|      | 289/289 [00:12<00:00, 23.88it/s, loss=23
Época 83/200 [Train] p=0.733  =8.67e-02: 100%|      | 289/289 [00:10<00:00, 26.45it/s, loss=23
Época 84/200 [Train] p=0.733  =8.67e-02: 100%|      | 289/289 [00:10<00:00, 26.75it/s, loss=23
Época 85/200 [Train] p=0.733  =8.67e-02: 100%|      | 289/289 [00:11<00:00, 26.18it/s, loss=23


 Época 85/200 | p: 0.733 |  : 8.67e-02
  Train - Total: 23.4312 | MSE: 1.4089 | L0: 445.3
  Val   - Total: 23.5794 | MSE: 1.4911 | L0: 447.5

Época 86/200 [Train] p=0.733  =8.67e-02: 100%|      | 289/289 [00:10<00:00, 26.74it/s, loss=23
Época 87/200 [Train] p=0.733  =8.67e-02: 100%|      | 289/289 [00:10<00:00, 26.64it/s, loss=23
Época 88/200 [Train] p=0.733  =8.67e-02: 100%|      | 289/289 [00:10<00:00, 26.70it/s, loss=22
Época 89/200 [Train] p=0.733  =8.67e-02: 100%|      | 289/289 [00:10<00:00, 26.62it/s, loss=23
Época 90/200 [Train] p=0.733  =8.67e-02: 100%|      | 289/289 [00:10<00:00, 26.73it/s, loss=23


 Época 90/200 | p: 0.733 |  : 8.67e-02
  Train - Total: 23.4091 | MSE: 1.3736 | L0: 443.5
  Val   - Total: 23.5673 | MSE: 1.4705 | L0: 444.8

Época 91/200 [Train] p=0.733  =8.67e-02: 100%|      | 289/289 [00:10<00:00, 26.68it/s, loss=23
Época 92/200 [Train] p=0.733  =8.67e-02: 100%|      | 289/289 [00:10<00:00, 26.90it/s, loss=23
Época 93/200 [Train] p=0.733  =8.67e-02: 100%|      | 289/289 [00:10<00:00, 26.92it/s, loss=23
Época 94/200 [Train] p=0.733  =8.67e-02: 100%|      | 289/289 [00:10<00:00, 26.70it/s, loss=23
Época 95/200 [Train] p=0.733  =8.67e-02: 100%|      | 289/289 [00:10<00:00, 26.83it/s, loss=22


 Época 95/200 | p: 0.733 |  : 8.67e-02
  Train - Total: 23.4076 | MSE: 1.3552 | L0: 443.0
  Val   - Total: 23.5585 | MSE: 1.4370 | L0: 445.1

Época 96/200 [Train] p=0.733  =8.67e-02: 100%|      | 289/289 [00:10<00:00, 26.61it/s, loss=23
Época 97/200 [Train] p=0.733  =8.67e-02: 100%|      | 289/289 [00:10<00:00, 26.64it/s, loss=23
Época 98/200 [Train] p=0.733  =8.67e-02: 100%|      | 289/289 [00:10<00:00, 26.75it/s, loss=22
Época 99/200 [Train] p=0.733  =8.67e-02: 100%|      | 289/289 [00:10<00:00, 27.00it/s, loss=23
Época 100/200 [Train] p=0.644  =8.33e-02: 100%|      | 289/289 [00:10<00:00, 26.55it/s, loss=2
```

```
 Época 100/200 | p: 0.644 |  : 8.33e-02
  Train - Total: 23.4548 | MSE: 1.4172 | L0: 439.7
  Val   - Total: 23.6035 | MSE: 1.4953 | L0: 441.7

Época 101/200 [Train] p=0.644  =8.33e-02: 100%|     | 289/289 [00:10<00:00, 26.98it/s, loss=2
Época 102/200 [Train] p=0.644  =8.33e-02: 100%|     | 289/289 [00:11<00:00, 26.21it/s, loss=2
Época 103/200 [Train] p=0.644  =8.33e-02: 100%|     | 289/289 [00:10<00:00, 26.81it/s, loss=2
Época 104/200 [Train] p=0.644  =8.33e-02: 100%|     | 289/289 [00:10<00:00, 26.61it/s, loss=2
Época 105/200 [Train] p=0.644  =8.33e-02: 100%|     | 289/289 [00:10<00:00, 26.90it/s, loss=2


 Época 105/200 | p: 0.644 |  : 8.33e-02
  Train - Total: 23.4429 | MSE: 1.3933 | L0: 439.6
  Val   - Total: 23.5940 | MSE: 1.4564 | L0: 442.4

Época 106/200 [Train] p=0.644  =8.33e-02: 100%|     | 289/289 [00:10<00:00, 26.79it/s, loss=2
Época 107/200 [Train] p=0.644  =8.33e-02: 100%|     | 289/289 [00:10<00:00, 27.00it/s, loss=2
Época 108/200 [Train] p=0.644  =8.33e-02: 100%|     | 289/289 [00:10<00:00, 26.77it/s, loss=2
Época 109/200 [Train] p=0.644  =8.33e-02: 100%|     | 289/289 [00:10<00:00, 26.54it/s, loss=2
Época 110/200 [Train] p=0.644  =8.33e-02: 100%|     | 289/289 [00:10<00:00, 26.45it/s, loss=2


 Época 110/200 | p: 0.644 |  : 8.33e-02
  Train - Total: 23.4380 | MSE: 1.3814 | L0: 439.6
  Val   - Total: 23.5929 | MSE: 1.4503 | L0: 441.7

Época 111/200 [Train] p=0.644  =8.33e-02: 100%|     | 289/289 [00:10<00:00, 26.84it/s, loss=2
Época 112/200 [Train] p=0.644  =8.33e-02: 100%|     | 289/289 [00:10<00:00, 26.28it/s, loss=2
Época 113/200 [Train] p=0.644  =8.33e-02: 100%|     | 289/289 [00:10<00:00, 26.83it/s, loss=2
Época 114/200 [Train] p=0.644  =8.33e-02: 100%|     | 289/289 [00:10<00:00, 26.83it/s, loss=2
Época 115/200 [Train] p=0.644  =8.33e-02: 100%|     | 289/289 [00:10<00:00, 26.52it/s, loss=2


 Época 115/200 | p: 0.644 |  : 8.33e-02
  Train - Total: 23.4328 | MSE: 1.3682 | L0: 439.6
  Val   - Total: 23.5810 | MSE: 1.4314 | L0: 442.4
```

```
Época 116/200 [Train] p=0.644  =8.33e-02: 100%|      | 289/289 [00:10<00:00, 26.96it/s, loss=2
Época 117/200 [Train] p=0.644  =8.33e-02: 100%|      | 289/289 [00:10<00:00, 26.68it/s, loss=2
Época 118/200 [Train] p=0.644  =8.33e-02: 100%|      | 289/289 [00:10<00:00, 26.69it/s, loss=2
Época 119/200 [Train] p=0.644  =8.33e-02: 100%|      | 289/289 [00:10<00:00, 26.73it/s, loss=2
Época 120/200 [Train] p=0.556  =7.95e-02: 100%|      | 289/289 [00:10<00:00, 26.88it/s, loss=2


 Época 120/200 | p: 0.556 |  : 7.95e-02
  Train - Total: 23.5746 | MSE: 1.5512 | L0: 436.2
  Val   - Total: 23.7259 | MSE: 1.6182 | L0: 438.2

Época 121/200 [Train] p=0.556  =7.95e-02: 100%|      | 289/289 [00:10<00:00, 26.58it/s, loss=2
Época 122/200 [Train] p=0.556  =7.95e-02: 100%|      | 289/289 [00:10<00:00, 26.93it/s, loss=2
Época 123/200 [Train] p=0.556  =7.95e-02: 100%|      | 289/289 [00:10<00:00, 26.70it/s, loss=2
Época 124/200 [Train] p=0.556  =7.95e-02: 100%|      | 289/289 [00:10<00:00, 26.99it/s, loss=2
Época 125/200 [Train] p=0.556  =7.95e-02: 100%|      | 289/289 [00:10<00:00, 26.63it/s, loss=2


 Época 125/200 | p: 0.556 |  : 7.95e-02
  Train - Total: 23.5301 | MSE: 1.5048 | L0: 436.0
  Val   - Total: 23.6917 | MSE: 1.5885 | L0: 438.2

Época 126/200 [Train] p=0.556  =7.95e-02: 100%|      | 289/289 [00:10<00:00, 26.66it/s, loss=2
Época 127/200 [Train] p=0.556  =7.95e-02: 100%|      | 289/289 [00:10<00:00, 26.76it/s, loss=2
Época 128/200 [Train] p=0.556  =7.95e-02: 100%|      | 289/289 [00:10<00:00, 26.83it/s, loss=2
Época 129/200 [Train] p=0.556  =7.95e-02: 100%|      | 289/289 [00:10<00:00, 26.79it/s, loss=2
Época 130/200 [Train] p=0.556  =7.95e-02: 100%|      | 289/289 [00:10<00:00, 26.87it/s, loss=2


 Época 130/200 | p: 0.556 |  : 7.95e-02
  Train - Total: 23.5109 | MSE: 1.4820 | L0: 435.9
  Val   - Total: 23.6486 | MSE: 1.5372 | L0: 438.2

Época 131/200 [Train] p=0.556  =7.95e-02: 100%|      | 289/289 [00:10<00:00, 26.93it/s, loss=2
Época 132/200 [Train] p=0.556  =7.95e-02: 100%|      | 289/289 [00:10<00:00, 26.97it/s, loss=2
Época 133/200 [Train] p=0.556  =7.95e-02: 100%|      | 289/289 [00:10<00:00, 26.90it/s, loss=2
Época 134/200 [Train] p=0.556  =7.95e-02: 100%|      | 289/289 [00:10<00:00, 26.75it/s, loss=2
Época 135/200 [Train] p=0.556  =7.95e-02: 100%|      | 289/289 [00:10<00:00, 26.85it/s, loss=2
```

```
 Época 135/200 | p: 0.556 |  : 7.95e-02
  Train - Total: 23.5071 | MSE: 1.4754 | L0: 435.9
  Val   - Total: 23.6593 | MSE: 1.5389 | L0: 438.4

Época 136/200 [Train] p=0.556  =7.95e-02: 100%|      | 289/289 [00:10<00:00, 26.66it/s, loss=2
Época 137/200 [Train] p=0.556  =7.95e-02: 100%|      | 289/289 [00:10<00:00, 27.05it/s, loss=2
Época 138/200 [Train] p=0.556  =7.95e-02: 100%|      | 289/289 [00:10<00:00, 26.70it/s, loss=2
Época 139/200 [Train] p=0.556  =7.95e-02: 100%|      | 289/289 [00:10<00:00, 27.07it/s, loss=2
Época 140/200 [Train] p=0.556  =7.95e-02: 100%|      | 289/289 [00:10<00:00, 26.67it/s, loss=2


 Época 140/200 | p: 0.467 |  : 7.56e-02
  Train - Total: 23.8613 | MSE: 1.8512 | L0: 433.3
  Val   - Total: 24.1282 | MSE: 2.0409 | L0: 434.4

Época 141/200 [Train] p=0.467  =7.56e-02: 100%|      | 289/289 [00:10<00:00, 26.99it/s, loss=2
Época 142/200 [Train] p=0.467  =7.56e-02: 100%|      | 289/289 [00:10<00:00, 26.83it/s, loss=2
Época 143/200 [Train] p=0.467  =7.56e-02: 100%|      | 289/289 [00:10<00:00, 26.87it/s, loss=2
Época 144/200 [Train] p=0.467  =7.56e-02: 100%|      | 289/289 [00:10<00:00, 26.74it/s, loss=2
Época 145/200 [Train] p=0.467  =7.56e-02: 100%|      | 289/289 [00:10<00:00, 26.82it/s, loss=2


 Época 145/200 | p: 0.467 |  : 7.56e-02
  Train - Total: 23.7786 | MSE: 1.7547 | L0: 432.8
  Val   - Total: 23.9516 | MSE: 1.8310 | L0: 435.3

Época 146/200 [Train] p=0.467  =7.56e-02: 100%|      | 289/289 [00:10<00:00, 26.71it/s, loss=2
Época 147/200 [Train] p=0.467  =7.56e-02: 100%|      | 289/289 [00:10<00:00, 27.06it/s, loss=2
Época 148/200 [Train] p=0.467  =7.56e-02: 100%|      | 289/289 [00:10<00:00, 26.78it/s, loss=2
Época 149/200 [Train] p=0.467  =7.56e-02: 100%|      | 289/289 [00:10<00:00, 27.18it/s, loss=2
Época 150/200 [Train] p=0.467  =7.56e-02: 100%|      | 289/289 [00:10<00:00, 26.58it/s, loss=2


 Época 150/200 | p: 0.467 |  : 7.56e-02
  Train - Total: 23.7580 | MSE: 1.7393 | L0: 432.8
  Val   - Total: 23.8805 | MSE: 1.7714 | L0: 434.7
```

```
Época 151/200 [Train] p=0.467  =7.56e-02: 100%|      | 289/289 [00:10<00:00, 27.03it/s, loss=2
Época 152/200 [Train] p=0.467  =7.56e-02: 100%|      | 289/289 [00:10<00:00, 27.00it/s, loss=2
Época 153/200 [Train] p=0.467  =7.56e-02: 100%|      | 289/289 [00:10<00:00, 26.75it/s, loss=2
Época 154/200 [Train] p=0.467  =7.56e-02: 100%|      | 289/289 [00:10<00:00, 26.97it/s, loss=2
Época 155/200 [Train] p=0.467  =7.56e-02: 100%|      | 289/289 [00:10<00:00, 26.81it/s, loss=2


 Época 155/200 | p: 0.467 |  : 7.56e-02
  Train - Total: 23.7291 | MSE: 1.7156 | L0: 432.8
  Val   - Total: 23.8880 | MSE: 1.7878 | L0: 434.8

Época 156/200 [Train] p=0.467  =7.56e-02: 100%|      | 289/289 [00:10<00:00, 26.97it/s, loss=2
Época 157/200 [Train] p=0.467  =7.56e-02: 100%|      | 289/289 [00:10<00:00, 26.89it/s, loss=2
Época 158/200 [Train] p=0.467  =7.56e-02: 100%|      | 289/289 [00:10<00:00, 27.04it/s, loss=2
Época 159/200 [Train] p=0.467  =7.56e-02: 100%|      | 289/289 [00:10<00:00, 26.93it/s, loss=2
Época 160/200 [Train] p=0.467  =7.56e-02: 100%|      | 289/289 [00:10<00:00, 26.96it/s, loss=2


 Época 160/200 | p: 0.378 |  : 7.13e-02
  Train - Total: 24.1003 | MSE: 2.0917 | L0: 431.4
  Val   - Total: 24.6730 | MSE: 2.5339 | L0: 431.9

Época 161/200 [Train] p=0.378  =7.13e-02: 100%|      | 289/289 [00:10<00:00, 26.65it/s, loss=2
Época 162/200 [Train] p=0.378  =7.13e-02: 100%|      | 289/289 [00:10<00:00, 26.91it/s, loss=2
Época 163/200 [Train] p=0.378  =7.13e-02: 100%|      | 289/289 [00:10<00:00, 26.68it/s, loss=2
Época 164/200 [Train] p=0.378  =7.13e-02: 100%|      | 289/289 [00:10<00:00, 26.76it/s, loss=2
Época 165/200 [Train] p=0.378  =7.13e-02: 100%|      | 289/289 [00:10<00:00, 26.59it/s, loss=2


 Época 165/200 | p: 0.378 |  : 7.13e-02
  Train - Total: 24.3150 | MSE: 2.2070 | L0: 430.8
  Val   - Total: 24.4754 | MSE: 2.2834 | L0: 432.3

Época 166/200 [Train] p=0.378  =7.13e-02: 100%|      | 289/289 [00:10<00:00, 26.82it/s, loss=2
Época 167/200 [Train] p=0.378  =7.13e-02: 100%|      | 289/289 [00:10<00:00, 26.90it/s, loss=2
Época 168/200 [Train] p=0.378  =7.13e-02: 100%|      | 289/289 [00:10<00:00, 27.08it/s, loss=2
Época 169/200 [Train] p=0.378  =7.13e-02: 100%|      | 289/289 [00:10<00:00, 27.02it/s, loss=2
Época 170/200 [Train] p=0.378  =7.13e-02: 100%|      | 289/289 [00:10<00:00, 26.83it/s, loss=2
```

```
 Época 170/200 | p: 0.378 |  : 7.13e-02
  Train - Total: 24.3050 | MSE: 2.1842 | L0: 431.0
  Val   - Total: 24.3060 | MSE: 2.1307 | L0: 432.2

Época 171/200 [Train] p=0.378  =7.13e-02: 100%|     | 289/289 [00:10<00:00, 26.94it/s, loss=2
Época 172/200 [Train] p=0.378  =7.13e-02: 100%|     | 289/289 [00:10<00:00, 26.84it/s, loss=2
Época 173/200 [Train] p=0.378  =7.13e-02: 100%|     | 289/289 [00:10<00:00, 26.92it/s, loss=2
Época 174/200 [Train] p=0.378  =7.13e-02: 100%|     | 289/289 [00:10<00:00, 26.77it/s, loss=2
Época 175/200 [Train] p=0.378  =7.13e-02: 100%|     | 289/289 [00:10<00:00, 26.94it/s, loss=2


 Época 175/200 | p: 0.378 |  : 7.13e-02
  Train - Total: 24.2470 | MSE: 2.1174 | L0: 431.2
  Val   - Total: 24.4697 | MSE: 2.2512 | L0: 432.8

Época 176/200 [Train] p=0.378  =7.13e-02: 100%|     | 289/289 [00:10<00:00, 26.52it/s, loss=2
Época 177/200 [Train] p=0.378  =7.13e-02: 100%|     | 289/289 [00:10<00:00, 26.90it/s, loss=2
Época 178/200 [Train] p=0.378  =7.13e-02: 100%|     | 289/289 [00:10<00:00, 26.68it/s, loss=2
Época 179/200 [Train] p=0.378  =7.13e-02: 100%|     | 289/289 [00:10<00:00, 26.90it/s, loss=2
Época 180/200 [Train] p=0.378  =7.13e-02: 100%|     | 289/289 [00:10<00:00, 26.66it/s, loss=2


 Época 180/200 | p: 0.289 |  : 6.69e-02
  Train - Total: 24.5115 | MSE: 2.3877 | L0: 430.9
  Val   - Total: 25.6362 | MSE: 3.3844 | L0: 431.0

Época 181/200 [Train] p=0.289  =6.69e-02: 100%|     | 289/289 [00:10<00:00, 26.97it/s, loss=2
Época 182/200 [Train] p=0.289  =6.69e-02: 100%|     | 289/289 [00:10<00:00, 26.69it/s, loss=2
Época 183/200 [Train] p=0.289  =6.69e-02: 100%|     | 289/289 [00:10<00:00, 26.96it/s, loss=2
Época 184/200 [Train] p=0.289  =6.69e-02: 100%|     | 289/289 [00:10<00:00, 26.54it/s, loss=2
Época 185/200 [Train] p=0.289  =6.69e-02: 100%|     | 289/289 [00:10<00:00, 27.06it/s, loss=2


 Época 185/200 | p: 0.289 |  : 6.69e-02
  Train - Total: 25.5706 | MSE: 2.9964 | L0: 432.3
  Val   - Total: 25.6019 | MSE: 2.9888 | L0: 433.2
```

```
Época 186/200 [Train] p=0.289  =6.69e-02: 100%|      | 289/289 [00:10<00:00, 26.73it/s, loss=2
Época 187/200 [Train] p=0.289  =6.69e-02: 100%|      | 289/289 [00:10<00:00, 26.84it/s, loss=2
Época 188/200 [Train] p=0.289  =6.69e-02: 100%|      | 289/289 [00:10<00:00, 26.97it/s, loss=2
Época 189/200 [Train] p=0.289  =6.69e-02: 100%|      | 289/289 [00:10<00:00, 26.78it/s, loss=2
Época 190/200 [Train] p=0.289  =6.69e-02: 100%|      | 289/289 [00:10<00:00, 26.98it/s, loss=2


 Época 190/200 | p: 0.289 |  : 6.69e-02
  Train - Total: 25.6538 | MSE: 3.0246 | L0: 433.1
  Val   - Total: 25.6897 | MSE: 2.9753 | L0: 434.4

Época 191/200 [Train] p=0.289  =6.69e-02: 100%|      | 289/289 [00:10<00:00, 26.83it/s, loss=2
Época 192/200 [Train] p=0.289  =6.69e-02: 100%|      | 289/289 [00:10<00:00, 26.87it/s, loss=2
Época 193/200 [Train] p=0.289  =6.69e-02: 100%|      | 289/289 [00:10<00:00, 26.64it/s, loss=2
Época 194/200 [Train] p=0.289  =6.69e-02: 100%|      | 289/289 [00:10<00:00, 26.84it/s, loss=2
Época 195/200 [Train] p=0.289  =6.69e-02: 100%|      | 289/289 [00:10<00:00, 26.75it/s, loss=2


 Época 195/200 | p: 0.289 |  : 6.69e-02
  Train - Total: 25.4105 | MSE: 2.8160 | L0: 433.3
  Val   - Total: 25.4662 | MSE: 2.8300 | L0: 434.0

Época 196/200 [Train] p=0.289  =6.69e-02: 100%|      | 289/289 [00:10<00:00, 26.97it/s, loss=2
Época 197/200 [Train] p=0.289  =6.69e-02: 100%|      | 289/289 [00:10<00:00, 26.79it/s, loss=2
Época 198/200 [Train] p=0.289  =6.69e-02: 100%|      | 289/289 [00:10<00:00, 26.96it/s, loss=2
Época 199/200 [Train] p=0.289  =6.69e-02: 100%|      | 289/289 [00:10<00:00, 26.68it/s, loss=2
Época 200/200 [Train] p=0.289  =6.69e-02: 100%|      | 289/289 [00:10<00:00, 27.01it/s, loss=2


 Época 200/200 | p: 0.289 |  : 6.69e-02
  Train - Total: 25.5176 | MSE: 2.8944 | L0: 433.5
  Val   - Total: 25.6995 | MSE: 3.0326 | L0: 434.1

======================================================================
 RESUMEN DEL ENTRENAMIENTO
======================================================================
Entrenamiento completo: 200 épocas

Métricas Finales (Última Época):
        Métrica   Train Validation
     Total Loss 25.5176    25.6995
```

```
       MSE Loss  2.8944     3.0326
    L0 Sparsity   433.5      434.1
Fraction Active   5.29%      5.30%
======================================================================
```

### 8. Guardar Modelo

```
# Guardar checkpoint final (local + organizado)
checkpoint = {
    'model_state_dict': model.state_dict(),
    'optimizer_state_dict': optimizer.state_dict(),
    'config': CONFIG,
    'history': history
}

# Guardado local
final_model_name = get_model_name(
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='final'
)
checkpoint_path = CONFIG['save_dir'] / final_model_name
torch.save(checkpoint, checkpoint_path)

# Guardado organizado
save_checkpoint(
    model,
    base_path=NAMING_META['base_path'],
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='final',
    extras=NAMING_META['extras'],
    config=CONFIG,
    history=history,
    optimizer=optimizer
)
```

```
# Resumen de archivos guardados
saved_files = pd.DataFrame({
    'Tipo': ['Local', 'Organizado'],
    'Ubicación': [
        str(checkpoint_path),
        f"layer_{NAMING_META['capa']:02d}/models/..."
    ]
})

print('\n' + '='*70)
print(' CHECKPOINTS GUARDADOS')
print('='*70)
print(saved_files.to_string(index=False))
print('='*70)
```

```
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
annealing_standard\12500games\sae_standard_p-annealing_l6_12500g_ex2_final.pt

======================================================================
 CHECKPOINTS GUARDADOS
======================================================================
      Tipo                                                  Ubicación
     Local saved_model\sae_standard_p-annealing_l6_12500g_final.pt
Organizado                                        layer_06/models/...
======================================================================
```

### 9. Métricas básicas del modelo

```
# Guardar metricas de entrenamiento
metrics_dir = get_metrics_dir(
    NAMING_META["experiment_base_path"],
    NAMING_META["variante"],
    NAMING_META["tecnica"],
    NAMING_META["capa"],
    NAMING_META["juegos"],
    extras_id=extras_id
)

training_metrics = {
    # Metricas de calidad
```

```
    "val_mse_final": history["val_mse_loss"][-1],
    "val_mse_best": best_val_mse,
    "val_l0_final": history["val_l0_sparsity"][-1],
    "val_fraction_active_final": history["val_fraction_active"][-1],
    # Contexto del entrenamiento
    "best_epoch": best_epoch,
    "total_epochs": len(history["train_total_loss"]),
    "train_mse_final": history["train_mse_loss"][-1],
    "train_l0_final": history["train_l0_sparsity"][-1],
    # Hiperparametros
    "hidden_dim": CONFIG["hidden_dim"],
    "expansion_factor": CONFIG["hidden_dim"] / CONFIG["input_dim"],
    "sparsity_coef": CONFIG["sparsity_coef"],
    "p_start": CONFIG["p_start"],
    "p_end": CONFIG["p_end"],
    "n_sparsity_updates": CONFIG["n_sparsity_updates"],
    "anneal_start": CONFIG["anneal_start"],
    "learning_rate": CONFIG["learning_rate"],
    "num_games": NAMING_META["juegos"],
}

metrics_path = metrics_dir / "training_metrics.json"
with open(metrics_path, "w") as f:
    _json.dump(training_metrics, f, indent=2)

metrics_df = pd.DataFrame({
    "Metrica": ["Val MSE (final)", "Val MSE (mejor)", "Val L0 (final)", "Fraccion Activa"],
    "Valor": [
        f"{training_metrics['val_mse_final']:.6f}",
        f"{training_metrics['val_mse_best']:.6f}",
        f"{training_metrics['val_l0_final']:.1f}",
        f"{training_metrics['val_fraction_active_final']:.2%}"
    ]
})
print("" + "="*50)
print(" METRICAS DE ENTRENAMIENTO GUARDADAS")
print("="*50)
print(metrics_df.to_string(index=False))
print(f"Archivo: {metrics_path}")
print("="*50)
```

```
==================================================
```

```
 METRICAS DE ENTRENAMIENTO GUARDADAS
==================================================
        Metrica    Valor
Val MSE (final) 3.032606
Val MSE (mejor)      inf
 Val L0 (final)    434.1
Fraccion Activa    5.30%
Archivo: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\
annealing_standard\12500games\ex2\metrics\training_metrics.json
==================================================
```

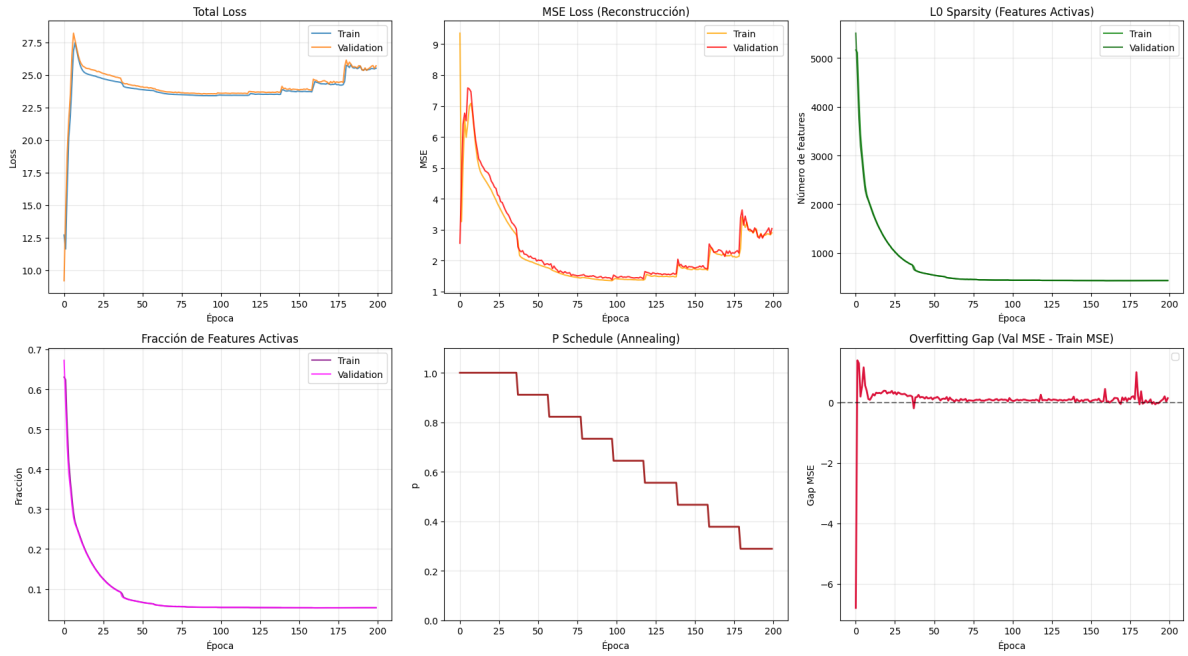
### 10. Visualización de Resultados

```python
# Gráficas de pérdidas (Train vs Validation) + P-annealing schedule
fig, axes = plt.subplots(2, 3, figsize=(18, 10))

# Total loss
axes[0, 0].plot(history['train_total_loss'], label='Train', alpha=0.8)
axes[0, 0].plot(history['val_total_loss'], label='Validation', alpha=0.8)
axes[0, 0].set_title('Total Loss')
axes[0, 0].set_xlabel('Época')
axes[0, 0].set_ylabel('Loss')
axes[0, 0].legend()
axes[0, 0].grid(True, alpha=0.3)

# MSE loss
axes[0, 1].plot(history['train_mse_loss'], label='Train', color='orange', alpha=0.8)
axes[0, 1].plot(history['val_mse_loss'], label='Validation', color='red', alpha=0.8)
if CONFIG['early_stopping'] and best_epoch > 0:
    axes[0, 1].axvline(best_epoch - 1, color='green', linestyle='--', alpha=0.7, label=f'Best
axes[0, 1].set_title('MSE Loss (Reconstrucción)')
axes[0, 1].set_xlabel('Época')
axes[0, 1].set_ylabel('MSE')
axes[0, 1].legend()
axes[0, 1].grid(True, alpha=0.3)

# L0 sparsity
axes[0, 2].plot(history['train_l0_sparsity'], label='Train', color='green', alpha=0.8)
axes[0, 2].plot(history['val_l0_sparsity'], label='Validation', color='darkgreen', alpha=0.8
axes[0, 2].set_title('L0 Sparsity (Features Activas)')
axes[0, 2].set_xlabel('Época')
```

```
axes[0, 2].set_ylabel('Número de features')
axes[0, 2].legend()
axes[0, 2].grid(True, alpha=0.3)

# Fraction active
axes[1, 0].plot(history['train_fraction_active'], label='Train', color='purple', alpha=0.8)
axes[1, 0].plot(history['val_fraction_active'], label='Validation', color='magenta', alpha=0
axes[1, 0].set_title('Fracción de Features Activas')
axes[1, 0].set_xlabel('Época')
axes[1, 0].set_ylabel('Fracción')
axes[1, 0].legend()
axes[1, 0].grid(True, alpha=0.3)

# P schedule (annealing)
axes[1, 1].plot(history['p_history'], color='brown', linewidth=2, label='p')
axes[1, 1].set_title('P Schedule (Annealing)')
axes[1, 1].set_xlabel('Época')
axes[1, 1].set_ylabel('p')
axes[1, 1].set_ylim(0, 1.1)
axes[1, 1].grid(True, alpha=0.3)

# Train vs Val MSE Gap
mse_gap = [v - t for v, t in zip(history['val_mse_loss'], history['train_mse_loss'])]
axes[1, 2].plot(mse_gap, color='crimson', linewidth=2)
axes[1, 2].axhline(0, color='black', linestyle='--', alpha=0.5)
if CONFIG['early_stopping'] and best_epoch > 0:
    axes[1, 2].axvline(best_epoch - 1, color='green', linestyle='--', alpha=0.7, label=f'Best
axes[1, 2].set_title('Overfitting Gap (Val MSE - Train MSE)')
axes[1, 2].set_xlabel('Época')
axes[1, 2].set_ylabel('Gap MSE')
axes[1, 2].legend()
axes[1, 2].grid(True, alpha=0.3)

plt.tight_layout()
plt.savefig(CONFIG['plots_dir'] / 'training_curves.png', dpi=300, bbox_inches='tight')
plt.show()

print('\n' + '='*50)
print('   Gráficas guardadas exitosamente')
print('='*50)
```

```
C:\Users\Esposa\AppData\Local\Temp\ipykernel_27156\2801488273.py:59: UserWarning: No artists
```

```
axes[1, 2].legend()
```




```
============================================================
   Gráficas guardadas exitosamente
============================================================
```

## 11. Resumen Final

```
# Resumen final completo
print('\n' + '='*70)
print(' RESUMEN FINAL DEL ENTRENAMIENTO')
print('='*70)

# Arquitectura
arch_summary = pd.DataFrame({
    'Componente': ['Input', 'Hidden', 'Parámetros'],
    'Valor': [
        f"{CONFIG['input_dim']}",
        f"{CONFIG['hidden_dim']} ({CONFIG['hidden_dim']/CONFIG['input_dim']:.1f}x)",
        f"{num_params:,}"
```

```
    ]
})
print('\nArquitectura:')
print(arch_summary.to_string(index=False))

# Datos
data_summary = pd.DataFrame({
    'Conjunto': ['Train', 'Validation', 'Total'],
    'Muestras': [
        f"{len(train_dataset):,}",
        f"{len(val_dataset):,}",
        f"{len(full_dataset):,}"
    ]
})
print('\nDatos:')
print(data_summary.to_string(index=False))

# Métricas finales
final_summary = pd.DataFrame({
    'Métrica': ['Train Loss', 'Val Loss', 'Train MSE', 'Val MSE', 'Train L0', 'Val L0'],
    'Valor': [
        f"{history['train_total_loss'][-1]:.4f}",
        f"{history['val_total_loss'][-1]:.4f}",
        f"{history['train_mse_loss'][-1]:.4f}",
        f"{history['val_mse_loss'][-1]:.4f}",
        f"{history['train_l0_sparsity'][-1]:.1f}",
        f"{history['val_l0_sparsity'][-1]:.1f}"
    ]
})
print('\nMétricas Finales:')
print(final_summary.to_string(index=False))

# Archivos
files_summary = pd.DataFrame({
    'Tipo': ['Modelo', 'Gráficas'],
    'Ubicación': [
        str(checkpoint_path),
        str(CONFIG['plots_dir'] / 'training_curves.png')
    ]
})
print('\nArchivos Guardados:')
print(files_summary.to_string(index=False))
```

```python
print('\n' + '='*70)
print('   ENTRENAMIENTO COMPLETADO EXITOSAMENTE')
print('='*70)
```

```
======================================================================
 RESUMEN FINAL DEL ENTRENAMIENTO
======================================================================

Arquitectura:
Componente        Valor
     Input          512
    Hidden 8192 (16.0x)
Parámetros    8,397,312

Datos:
  Conjunto Muestras
     Train  590,000
Validation  147,500
     Total  737,500

Métricas Finales:
   Métrica   Valor
Train Loss 25.5176
  Val Loss 25.6995
 Train MSE  2.8944
   Val MSE  3.0326
  Train L0   433.5
    Val L0   434.1

Archivos Guardados:
    Tipo
  Modelo
annealing_l6_12500g_final.pt
Gráficas c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\
annealing_standard\12500games\ex2\plots\training_curves.png

======================================================================
   ENTRENAMIENTO COMPLETADO EXITOSAMENTE
======================================================================
```

### 12. Generar PDF con Quarto

Una vez ejecutado todo el notebook, ejecutar el siguiente comando en la terminal para generar el PDF con Quarto:

```
import subprocess

# Obtener directorio del experimento y nombre del PDF
exp_dir = get_experiment_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)

pdf_name = get_report_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    'training',
    extras_id=extras_id
)

notebook_name = 'train_sae_standard_pannealing.ipynb'
output_path = exp_dir / pdf_name

print(f' Generando PDF con Quarto...')
print(f' Archivo de salida: {output_path}')

# Quarto no acepta rutas en --output, solo nombre de archivo
# Se usa --output-dir para el directorio y --output solo para el nombre
result = subprocess.run(
    f'quarto render {notebook_name} --to pdf --output-dir "{exp_dir}" --output {pdf_name}',
    shell=True,
    capture_output=True,
    text=True
)

if result.returncode == 0:
    print(f' PDF generado exitosamente: {output_path}')
```

```
else:
    print(f' Error al generar PDF:')
    print(result.stderr)
```

```
 Generando PDF con Quarto...
 Archivo de salida: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sa
annealing_standard\12500games\ex2\sae_standard_p-annealing_l6_12500g_ex2_training.pdf
 PDF generado exitosamente: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\lay
annealing_standard\12500games\ex2\sae_standard_p-annealing_l6_12500g_ex2_training.pdf
```